SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct (192325060)

I H.Mohammed Nawaz (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Nawaz

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Q1) Cloud Auto-scaling and load Balancing:-

Introduction:- cloud auto-scaling and load balancing help organizations handle sudden increases in website traffic without affecting performance.

Cloud Auto-scaling:-

* Automatically adds or removes virtual servers based on user demand

* prevents server overload during peak traffic.

* Reduces infrastructure cost by removing unused resources.

Load Balancing:-

* Distributes incoming requests across multiples servers

* prevents a singe server from becoming overloade

* Improves application availability and reliability.

Role of monitoring tools :-

* monitors cpu, memory, network traffic, and response time.

Benefits:-

* Reduces downtime.

* improves websites speed

* provides better user experience.

conclusion:- Amazon and flipkart use auto-scaling and load balancing during festive sales like Big Billion Days to manage millions of users without website crashes.

Q2) cloud storage for secure Backups:

Introduction:- cloud storage allows organizations to store large amounts of backups data securely while ensuring high availability.

Data Durability and Redundancy:

* storage multiple copies of data.

* Replicates data across different servers and regions

* prevents data loss even if one server fails.

Security measures:-

* Encryption:- protects data during storage.

* Access control

* Multi-Factor Authentication (MFA).

Cost Advantages:-

* No need to purchase expensive storage hardware

* pay only for the storage used.

* Lower maintenance & operational costs.

Disaster recovery Example:-

A hospital stores patients records in the cloud. If the local server is damaged due to fire or flooding, the backup can be restored quickly from the cloud.

Q3) Content Delivery Networks (CDNs):-

Introduction:- A content delivery Network (CND) Improves websites and application performance by delivering content from server located closer to users

How CDN Improve performance

*Reduces loading time.

*Delivers content from the nearest edge server.

*Decreases network congestion.

Roles of edges servers:-

*Store frequently accessed content.

* Reduce latency.

*Improve response time for users worldwide.

cloud provider support:-

*offers global data centers.

*provides high availability and reliability.

*ensures continuous service even during high result.

Benefits:-

*Faster websites access.

*Better user experience.

Real world Example: Netflix uses CDNS to stream high quality videos without buffering, while online games use CDNS to reduce gameplay lag.

Q4) Cloud security and Regulatory compliance.

Introduction:-

Financial institutions require secure cloud environment to protect customer data and comply with government regulations.

Regulatory compliance:-

* supports standards such as PCI, DSS, ISO 27001, and GDPR.
* performs regular security audits.
* maintains compliance reports.

Role of logging, monitoring and encryption.

* Logging: Records all user and system activities.
* monitoring:- Detects suspicious activities in realtime
* Encryption:- protects sensitives financial information

Real-world Example:

Banks encrypt online transaction separates services such as login, product catalog payments, orders and Notification into Individual containers.

Q5) Containers and Kubernetes:-

containers provide a lightweight method of deploying application while Kubernetes automates their management and scaling

Containers (vs) Virtual machines:-

*containers share the host operating system.
*use fewer systems resources.

Role of Kubernetes:-

*Automates application deployment.
*performs load balancing.
* Restarts failed containers.

portability Benefits:-

* Applications run consistently on AWS, Azure.
* Easy migration between cloud providers.
*Reduces compatibility issues.

conclusion:-

An e-commerce application separates service such as login, product, catalog, payments, orders, and notifications into individual containers.